

Formal verification of the consistency of high order schemes for the anisotropic divergence form Laplacian

This notebook is companion code for the manuscript :

“Anisotropic wave equation and Selling’s decomposition”, by Malik Halaoui, Ludovic Métivier, Jean-Marie Mirebeau

Finite differences

The following expressions obey the Taylor expansion $\langle e, \text{grad } u(x) \rangle + O(|e|^{k+1})$, where k is the accuracy order.

They are used to construct the Selling schemes.

```
In[*]:= (*Upwind, first order accurate*)
Diff1u[u_, e_][x_] := u[x + e] - u[x];
(*Centered, second order accurate*)
Diff2c[u_, e_][x_] := (u[x + e] - u[x - e]) / 2;
(*Upwind, second order accurate*)
Diff2u[u_, e_][x_] := (-u[x + 2 e] + 4 u[x + e] - 3 u[x]) / 2;
(*Semi-centered-upwind, third order accurate*)
Diff3s[u_, e_][x_] := (-u[x + 2 e] + 6 u[x + e] - 3 u[x] - 2 u[x - e]) / 6;
(*Upwind, third order accurate*)
Diff3u[u_, e_][x_] := (2 u[x + 3 e] - 9 u[x + 2 e] + 18 u[x + e] - 11 u[x]) / 6;
```

Let us verify the announced consistency order.

```
In[*]:= Series[Diff1u[u, h][x], {h, 0, 2}]
Out[*]=
```

$$u'[x] h + \frac{1}{2} u''[x] h^2 + O[h]^3$$

```
In[*]:= Series[Diff2c[u, h][x], {h, 0, 3}]
Out[*]=
```

$$u'[x] h + \frac{1}{6} u^{(3)}[x] h^3 + O[h]^4$$

```
In[*]:= Series[Diff2u[u, h][x], {h, 0, 3}]
Out[*]=
```

$$u'[x] h - \frac{1}{3} u^{(3)}[x] h^3 + O[h]^4$$

```
In[*]:= Series[Diff3s[u, h][x], {h, 0, 4}]
Out[*]=
```

$$u'[x] h - \frac{1}{12} u^{(4)}[x] h^4 + O[h]^5$$

```
In[*]:= Series[Diff3u[u, h][x], {h, 0, 4}]
Out[*]=
```

$$u'[x] h + \frac{1}{4} u^{(4)}[x] h^4 + O[h]^5$$

The following expressions are used to construct the high order variants of the axis-aligned and criss-cross schemes.

```
In[*]:= (*Centered, fourth order accurate*)
Diff4c[u_, e_][x_] := (-u[x + 2 e] + 8 u[x + e] - 8 u[x - e] + u[x - 2 e]) / 12;
(*Semi-centered-upwind, fourth order consistent with <e, grad u(x+e/2)>*)
Diff4s[u_, e_][x_] := (-u[x + 2 e] + 27 u[x + e] - 27 u[x] + u[x - e]) / 24;
(*Semi-centered-upwind, interpolation operator,
fourth order consistent with u(x+e/2)*)
Interp4s[u_, e_][x_] := (-u[x + 2 e] + 9 u[x + e] + 9 u[x] - u[x - e]) / 16;
```

Again, let us verify the announced consistency orders .

```
In[*]:= Series[Diff4c[u, h][x], {h, 0, 6}]
Out[*]=
```

$$u'[x] h - \frac{1}{30} u^{(5)}[x] h^5 + O[h]^7$$

```
In[*]:= Series[Diff4s[u, h][x], {h, 0, 5}] - u'[x + h/2] h
Out[*]=
```

$$-\frac{3}{640} u^{(5)}[x] h^5 + O[h]^6$$

```
In[*]:= Series[Interp4s[u, h][x], {h, 0, 4}] - u[x + h/2]
Out[*]=
```

$$-\frac{3}{128} u^{(4)}[x] h^4 + O[h]^5$$

Consistency error of the squared derivative

We now build approximations of $\langle e, \text{grad } u(x) \rangle^2 + O(|e|^{2+k})$, where k is the consistency order.

```
In[*]:= (*Second order accurate*)
Series[(Diff1u[u, -h][x]^2 + Diff1u[u, h][x]^2) / 2, {h, 0, 4}]
```

```
Out[*]=
```

$$u'[x]^2 h^2 + \frac{1}{12} (3 u''[x]^2 + 4 u'[x] u^{(3)}[x]) h^4 + O[h]^5$$

In[*]:= (*Fourth order accurate*)

Series[

(Diff2u[u, -h][x]^2 + 4 Diff2c[u, h][x]^2 + Diff2u[u, h][x]^2) / 6, {h, 0, 6}]

Out[*]=

$$u'[x]^2 h^2 + \frac{1}{90} (5 u^{(3)}[x]^2 - 6 u'[x] u^{(5)}[x]) h^6 + O[h]^7$$

In[*]:= (*Sixth order accurate*)

Series[(Diff3u[u, -h][x]^2 + 9 Diff3s[u, -h][x]^2 +

9 Diff3s[u, h][x]^2 + Diff3u[u, h][x]^2) / 20, {h, 0, 8}]

Out[*]=

$$u'[x]^2 h^2 + \frac{1}{20} \left(\frac{1}{2} \left(\frac{1}{4} u^{(4)}[x]^2 - \frac{2}{7} u'[x] u^{(7)}[x] \right) + \frac{1}{18} \left(\frac{9}{4} u^{(4)}[x]^2 + \frac{54}{7} u'[x] u^{(7)}[x] \right) \right) h^8 + O[h]^9$$

Similarly, we obtain approximations of $-(\lambda u)'$, as required for the consistency proof of the Selling schemes.

These computations, for high order schemes, are mentioned in the end of section 3.2.

(*Second order accurate*)

Series[(Diff1u[λ[#] × Diff1u[u, h][#] &, -h][x] +

Diff1u[λ[#] × Diff1u[u, -h][#] &, h][x]) / 2, {h, 0, 4}]

Out[*]=

$$(-u'[x] \lambda'[x] - \lambda[x] u''[x]) h^2 + \frac{1}{12} (-3 u''[x] \lambda''[x] - 2 \lambda'[x] u^{(3)}[x] - 2 u'[x] \lambda^{(3)}[x] - \lambda[x] u^{(4)}[x]) h^4 + O[h]^5$$

(*Fourth order accurate*)

In[*]:= Series[(Diff2u[λ[#] × Diff2u[u, h][#] &, -h][x] +

4 Diff2c[λ[#] × Diff2c[u, -h][#] &, h][x] +

Diff2u[λ[#] × Diff2u[u, -h][#] &, h][x]) / 6, {h, 0, 6}]

Out[*]=

$$(-u'[x] \lambda'[x] - \lambda[x] u''[x]) h^2 + \frac{1}{90} (25 u^{(3)}[x] \lambda^{(3)}[x] + 15 \lambda''[x] u^{(4)}[x] + 15 u''[x] \lambda^{(4)}[x] + 3 \lambda'[x] u^{(5)}[x] + 3 u'[x] \lambda^{(5)}[x] + \lambda[x] u^{(6)}[x]) h^6 + O[h]^7$$

(*Sixth order accurate*)

In[*]:= Simplify@Series[

(Diff3u[λ[#] × Diff3u[u, h][#] &, -h][x] + 9 Diff3s[λ[#] × Diff3s[u, h][#] &, -h][

x] + 9 Diff3s[λ[#] × Diff3s[u, -h][#] &, h][x] +

Diff3u[λ[#] × Diff3u[u, -h][#] &, h][x]) / 20, {h, 0, 8}]

Out[*]=

$$(-u'[x] \lambda'[x] - \lambda[x] u''[x]) h^2 + \frac{1}{560} (-133 u^{(4)}[x] \lambda^{(4)}[x] - 112 \lambda^{(3)}[x] u^{(5)}[x] - 84 u^{(3)}[x] \lambda^{(5)}[x] - 42 \lambda''[x] u^{(6)}[x] - 28 u''[x] \lambda^{(6)}[x] - 4 \lambda'[x] u^{(7)}[x] - 4 u'[x] \lambda^{(7)}[x] - \lambda[x] u^{(8)}[x]) h^8 + O[h]^9$$

Dispersion relation

The dispersion relation is obtained by considering planewaves and constant PDE coefficients. These computations, for high order schemes, are mentioned in the end of section 3.3.

(*Second order scheme*)

```
ExpToTrig[
  (Diff1u[Diff1u[u, h] [#] &, -h] [0] + Diff1u[Diff1u[u, -h] [#] &, h] [0]) / 2 /.
  u[x_] → Exp[I x]]
```

Out[*]=

$$2 - 2 \cos[h]$$

(*Fourth order scheme*)

```
ExpToTrig[
  (Diff2u[Diff2u[u, h] [#] &, -h] [0] + 4 Diff2c[Diff2c[u, -h] [#] &, h] [0] +
  Diff2u[Diff2u[u, -h] [#] &, h] [0]) / 6 /. u[x_] → Exp[I x]]
```

Out[*]=

$$\frac{5}{2} - \frac{8 \cos[h]}{3} + \frac{1}{6} \cos[2h]$$

```
In[*]:= Series[ $\frac{5}{2} - \frac{8 \cos[h]}{3} + \frac{1}{6} \cos[2h]$ , {h, 0, 7}]
```

Out[*]=

$$h^2 - \frac{h^6}{90} + O[h]^8$$

(*Sixth order scheme*)

```
ExpToTrig[
  (Diff3u[Diff3u[u, h] [#] &, -h] [0] + 9 Diff3s[Diff3s[u, h] [#] &, -h] [0] +
  9 Diff3s[Diff3s[u, -h] [#] &, h] [0] +
  Diff3u[Diff3u[u, -h] [#] &, h] [0]) / 20 /. u[x_] → Exp[I x]]
```

Out[*]=

$$\frac{49}{18} - 3 \cos[h] + \frac{3}{10} \cos[2h] - \frac{1}{45} \cos[3h]$$

```
In[*]:= Series[ $\frac{49}{18} - 3 \cos[h] + \frac{3}{10} \cos[2h] - \frac{1}{45} \cos[3h]$ , {h, 0, 9}]
```

Out[*]=

$$h^2 - \frac{h^8}{560} + O[h]^{10}$$